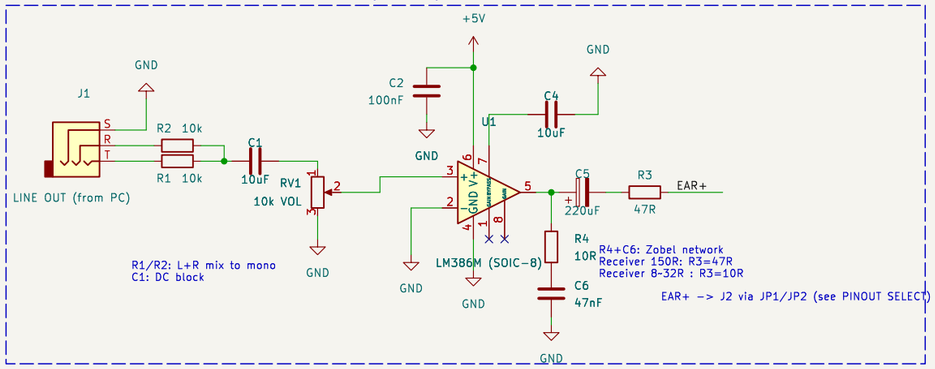
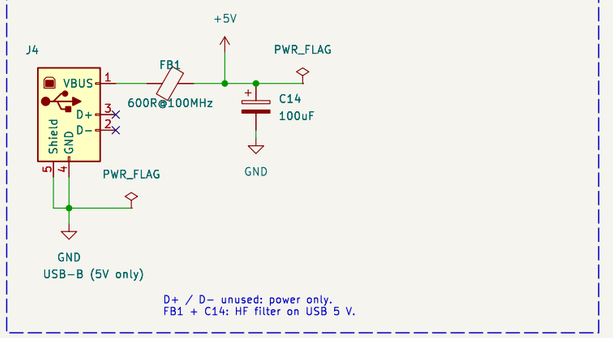
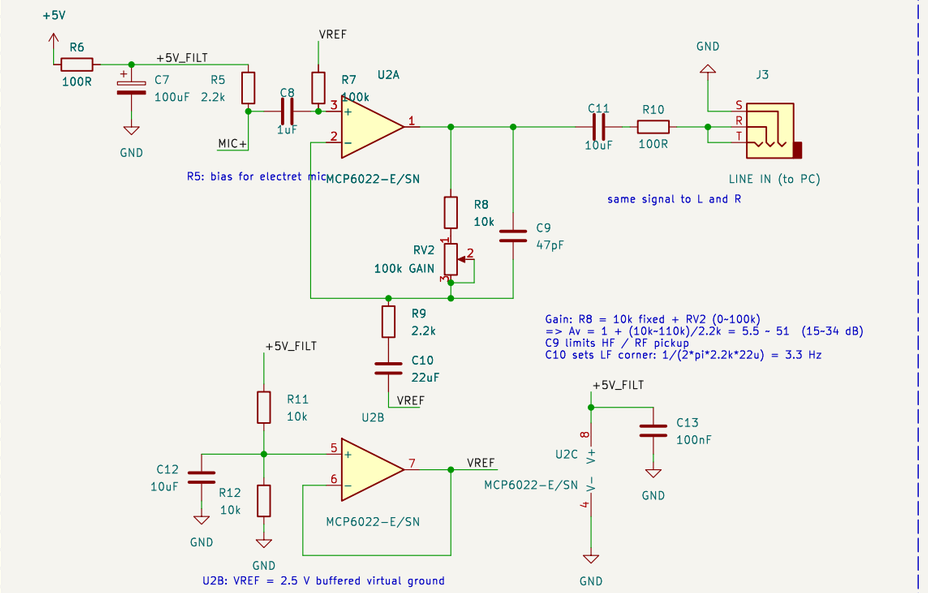
## PATH A : PC LINE OUT -> LM386 (Av=20) -> Handset receiver　路徑 A:電腦 line out → 聽筒




## POWER : 5 V from PC USB (shares PC ground -> no ground loop)　電源與接地




## PATH B : MIC -> MCP6022 preamp (Av 5.5~51) -> PC LINE IN　路徑 B:麥克風 → line in

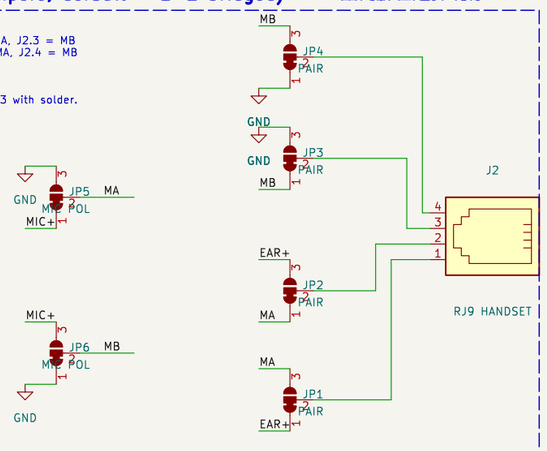



## NOTES　設計說明

1. RJ9 (4P4C) pinout is NOT standardized. Measure first:
   - receiver pair: 30-300 ohm DC; electret mic pair: diode-like (asymmetric).
   - set JP1-JP4 (pair) and JP5-JP6 (mic polarity) to match the handset.
2. Path A: LM386 has low output impedance, so the receiver no longer loads the PC line out -> no volume drop. RV1 sets the level. Gain fixed at 20 (pins 1/8 open). C5 blocks the 2.5 V DC offset. R3 limits current into the receiver (telephone receiver ~1 mW rated).
3. Path B: electret needs 2-5 V bias via R5. Mic level ~10 mV -> line level ~300 mV needs ~30 dB; set with RV2. Output impedance ~100 ohm drives the 10 kohm line in without loss. Dynamic mic instead of electret: remove R5, raise gain (2 stages).
4. Ground: everything is powered from the same PC USB port -> single ground reference. Star ground on PCB; keep U2 away from U1 output. R6/C7 filter USB switching noise for the mic preamp (+5V_FILT).
5. Optional sidetone: 47k-100k from U2A output to RV1 pin 1.
6. Optional LDO (e.g. 4.5 V) in place of R6 if USB noise is still audible.
7. SMT: R/C 0805; 10uF/22uF = 1206 X5R 10V ceramic; 47pF = C0G; U1/U2 = SOIC-8; RV2 = Bourns 3314J SMT trimmer; FB1 = 0805 bead. THT (connectors + panel pot): J1/J3 = CUI SJ1-3523N 3.5 mm jack, J2 = Evercom 5301-440 RJ9, J4 = Lumberg 2411 02 USB-B, RV1 = Alps RK097 single 9 mm right-angle shaft pot.
8. C5/C7/C14 = SMD V-chip electrolytic 6.3 mm, >= 6.3 V rating.

## J2 HANDSET PINOUT SELECT (solder jumpers, default = 1-2 bridged)　話筒腳位選擇跳線

JP1-JP4 = PAIR select (always set all four the same way)
  1-2 (default): J2.1 = EAR+, J2.4 = EAR-/GND, J2.2 = MA, J2.3 = MB
  2-3          : J2.2 = EAR+, J2.3 = EAR-/GND, J2.1 = MA, J2.4 = MB
JP5-JP6 = MIC polarity (always set both the same way)
  1-2 (default): MA = MIC+ (hot), MB = GND
  2-3          : MA = GND, MB = MIC+ (hot)
To change: cut the 1-2 trace on the jumper, then bridge 2-3 with solder.




| | |
|---|---|
| Sheet: / | |
| File: handset_iface.kicad_sch | |
| **Title: RJ9 Handset <-> PC Line In / Line Out Interface** | |
| Size: A3 | Date: 2026-09-28 |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 1/1 |